Contact-Rich Slide–Pick Manipulation Planning for Dense Cuboid Clutter with Parallel Grippers

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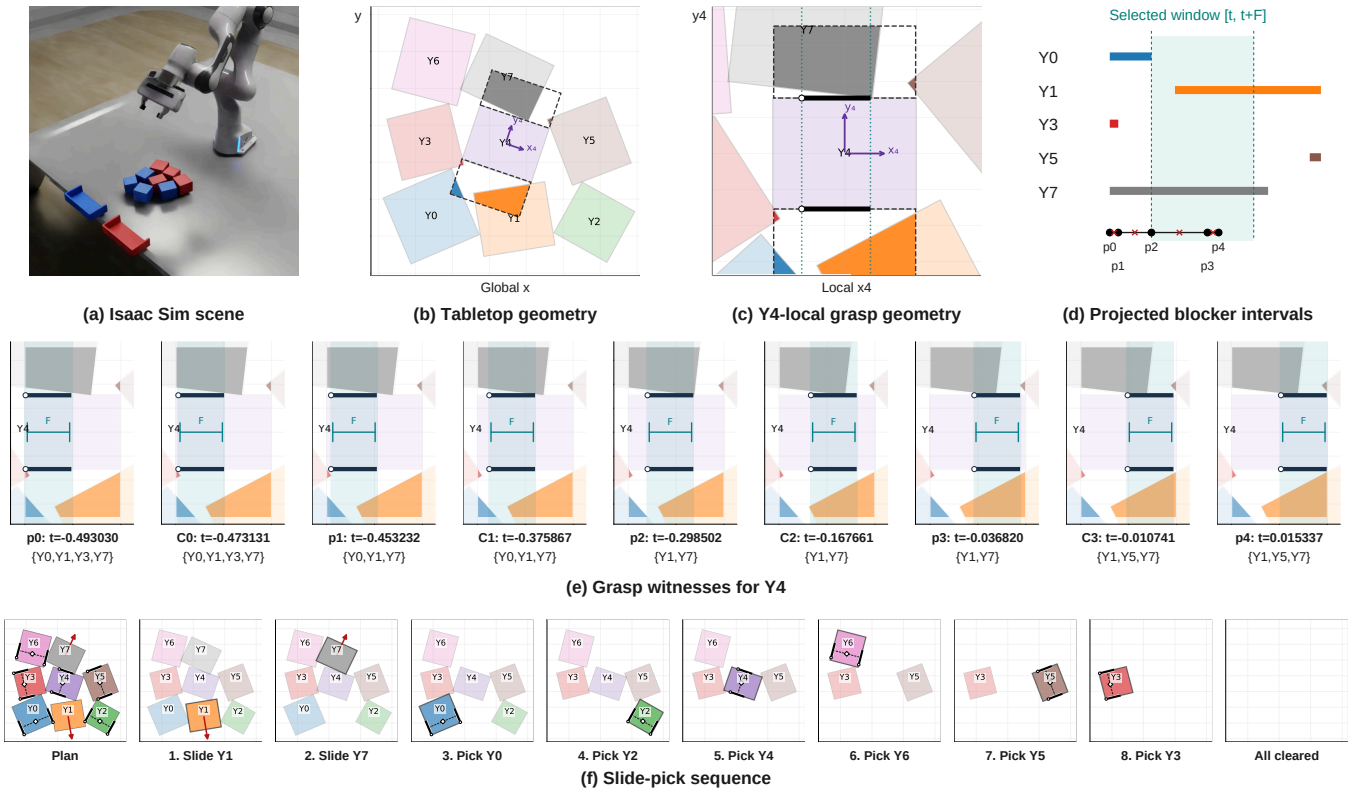


Fig. 1: Slide-pick planning. (a) A tabletop sorting scene where a parallel gripper places each cuboid into its matching-color chute. (b) Tabletop footprints. (c) Y4-local finger windows. (d) Projected intervals yield five event points and four open-cell representatives. (e) Grasp witnesses and blocker sets. (f) Two slides and eight picks. Red arrows indicate slide directions, black bars finger-contact windows, diamonds grasp centers, and circles window starts.

***Abstract*— Manipulation of tightly packed tabletop cuboids with parallel grippers becomes challenging when neighboring objects eliminate the clearance required for a feasible grasp, necessitating non-prehensile interaction. We study how directional sliding can create grasp clearance for heterogeneous cuboids in axis-aligned and non-axis-aligned arrangements. Our task-planning formulation combines continuously parameterized picks with slides and minimizes the number of slides required to clear the scene. For an object–orientation pair with m projected blocking intervals, we show that at most $4m + 3$ witnesses preserve every grasp blocker set, and construct the event decomposition in $O(m \log m)$ time. We derive an exact finite representation of grasp feasibility from projected blocking intervals. The resulting dependencies form a monotone Horn system with order-independent pick closure. Minimum-slide planning is characterized as a minimum-key problem, with NP-completeness established for the abstract rule system and an exact mixed-integer linear program for the geometric instances. We evaluate the planner on 40 benchmark scenes with 2,000 physics-simulated executions in Isaac Sim, where our method outperforms all four baselines. We further validate the integrated perception–planning–execution pipeline through real-robot experiments, with all trials successfully completed. Online pose updates and slide reattempts address insufficient grasp clearance during execution. The results demonstrate how geometric task planning and perception feedback can support the composition of prehensile and non-prehensile actions in dense tabletop clutter.**

I. Introduction and Related Work

Warehouse automation, logistics, and related manipulation applications frequently involve boxes arranged on planar work surfaces. Parallel grippers, despite being widely used end-effectors, face challenges in densely packed arrangements. Neighboring objects may eliminate the opposing-finger clearance necessary for a top-down grasp, even when each object is graspable in isolation. Such scenes may therefore require manipulation beyond prehensile pick-and-place, using non-prehensile actions to create clearance [1], [2]. This work considers directional sliding as one such primitive and studies how to combine it with picking while minimizing contact actions.

The geometric dependencies become more complex when cuboids have different dimensions, partial side contacts, and arbitrary in-plane orientations. A parallel grasp can be continuously offset along an object side, and different placements may require different neighboring objects to be removed. Adjacency alone therefore does not determine pickability. A neighboring object may obstruct one grasp placement while leaving another feasible. Conversely, a single placement may require several blockers to be cleared together. The planning representation must preserve both the continuous grasp choices and their conjunctive dependencies.

Non-prehensile manipulation has been studied through pushing, tapping, and frictional contact mechanics [3]–[5], including push–grasping in clutter [1], [2]. Task and motion planning integrates discrete action choices with continuous geometric constraints [6]–[8]. Related sequencing problems arise in rearrangement [9]–[11], dense packing [12], [13], and assembly planning [14]–[16]. Here, we focus on the geometric structure of parallel-jaw grasping and directional sliding, separating their high-level dependencies from the low-level contact controller.

We represent continuous finger placement through a finite set of witnesses that preserves every grasp blocker set. These options become definite Horn implications, or directed hyperedges with conjunctive predecessor sets [17], [18]. Together with slide-corridor dependencies, they define a monotone system related to delete-free planning [19]–[21].

Our contributions are an exact $4m+3$ grasp-witness bound with an $O(m \log m)$ event construction, order-independent pick closure, and a minimum-key characterization of slide planning. We establish NP-completeness for the abstract rule system through Target Set Selection [22] and derive an exact MILP for geometric instances. Simulation and physical experiments evaluate planned slides, contact attempts, execution time, and completion.

Figure 1 illustrates the formulation. The cluttered scene and its footprints are shown in Fig. 1(a)–(b). For object Y4, the opposed finger windows in its local frame induce the projected blocker intervals and finite grasp witnesses in Fig. 1(c)–(e). The sequence in Fig. 1(f) selects Y1 and Y7 for sliding before their picks, followed by six direct picks. During execution, perception updates object poses and additional slide attempts are issued when the observed clearance is insufficient. This connects the planned choice of contact action to the geometry required by the subsequent grasp.

II. Problem Formulation and Method

A. Scene and manipulation model

Let $V = \{1, \dots, n\}$ index cuboids with pairwise interior-disjoint rectangular footprints

$$P_i = c_i + Q_i([-w_i/2, w_i/2] \times [-h_i/2, h_i/2]), \quad (1)$$

where Q_i is the planar rotation. Geometry relative to object i is expressed in its local frame as $P_{j|i} = Q_i^\top (P_j - c_i)$. Partial contacts are allowed. A state $R \subseteq V$ records objects permanently cleared from the packed scene, while uncleared objects remain at their initial footprints. We assume that isolated objects are reachable and graspable, and that destination placements do not obstruct later actions.

A *Pick* directly removes an object. A *Slide* translates it along a local axis to create clearance, followed by its required pick/place. Each selected slide assumes a realizable contact approach and a collision-free translation to staging space. Every object incurs one mandatory pick/place, so a complete plan Π has $n + C_S(\Pi)$ nominal actions. We minimize $\mathrm{OPT}_S = \min_\Pi C_S(\Pi)$, while physical retries are measured separately.

B. Finite parallel-jaw grasp rules

Let $F > 0$ be the tangential finger-contact length, $D > 0$ the maximum opening, and $\eta_D, \eta_F \geq 0$ optional opening and normal-clearance margins. The admissible local closing axes are

$$\begin{aligned} \mathcal{O}_i = &\{x : w_i + \eta_D \leq D,\ h_i \geq F\} \\ &\cup \{y : h_i + \eta_D \leq D,\ w_i \geq F\}. \end{aligned} \quad (2)$$

For x-closing, the finger-window start lies in $T_i^x = [-h_i/2, h_i/2 - F]$. The y-closing domain exchanges w_i and h_i. Project neighboring geometry within each opposed side's clearance strip onto the tangential coordinate, using side-contact intervals when $\eta_F = 0$. Let $B_i^o(t)$ contain the objects whose intervals positively overlap $[t, t+F]$. The grasp is feasible at R exactly when $B_i^o(t) \subseteq R$.

Exact grasp discretization. A window positively overlaps a projected interval $[a, b]$ precisely when $a - F < t < b$. Each interval therefore contributes two events, $a - F$ and b. With $m = m_i^o$ intervals, these events and the domain endpoints partition the domain into at most $2m + 2$ event points and $2m + 1$ open cells. Retaining every event point and one representative per cell yields

$$\begin{aligned} &|\mathcal{S}_i^o| \leq 4m + 3, \\ &\exists t \in T_i^o : B_i^o(t) \subseteq R \iff \exists \hat{t} \in \mathcal{S}_i^o : B_i^o(\hat{t}) \subseteq R. \end{aligned} \quad (3)$$

Blocker membership is constant within each open cell. Event points must also be retained because positive overlap excludes endpoint-only contact. Sorting and sweeping the events takes $O(m \log m)$ time, with additional cost proportional to the explicitly materialized blocker memberships.

Duplicate blocker sets are merged, and a grasp g_1 is discarded when $B(g_1) \supset B(g_2)$. The retained set $\mathcal{G}_i$ satisfies

$$i \in \mathrm{Pick}(R) \iff \bigvee_{g \in \mathcal{G}_i} \bigwedge_{j \in B(g)} [j \in R]. \quad (4)$$

Each grasp is a zero-cost Horn rule $B(g) \to i$. In Fig. 1(e), the nine witnesses produce four distinct blocker sets. Dominance pruning retains a witness requiring only Y1 and Y7 to be cleared.

C. Slide rules and minimum-key planning

For a local direction $d \in \{\pm e_x, \pm e_y\}$ and a certified distance $\delta_i(d) > 0$, define

$$\begin{aligned} W_i(d) &= \bigcup_{0 < \tau \leq \delta_i(d)} \mathrm{int}(P_i + \tau Q_i d), \\ K(i,d) &= \{j \neq i : \mathrm{int}(P_j) \cap W_i(d) \neq \varnothing\}. \end{aligned} \quad (5)$$

Category	Method	T_p (ms)	Total actions	Contact actions	Contact attempts	T_e (s)	Success (%)
Axis-aligned	Proposed	25.04 ± 17.03	$\mathbf{8.85 \pm 2.78}$	$\mathbf{1.55 \pm 1.05}$	$\mathbf{2.35 \pm 2.03}$	$\mathbf{60.73 \pm 24.83}$	**100**
	Greedy Closure	15.83 ± 7.65	8.95 ± 2.84	1.65 ± 1.14	3.85 ± 5.85	67.49 ± 30.98	**100**
	Greedy Min-Blockers	$\mathbf{15.82 \pm 7.64}$	10.00 ± 3.29	2.70 ± 1.84	3.80 ± 3.56	66.92 ± 28.80	**100**
	Random Slide	16.39 ± 7.86	9.71 ± 3.27	2.42 ± 1.56	3.77 ± 3.03	72.92 ± 31.93	**100**
	Random Push	247.92 ± 190.19	49.16 ± 32.87	18.80 ± 13.93	18.80 ± 13.93	197.01 ± 127.89	65.5
Non-axis-aligned	Proposed	27.48 ± 25.55	$\mathbf{8.75 \pm 2.51}$	$\mathbf{1.25 \pm 0.55}$	$\mathbf{1.70 \pm 1.03}$	$\mathbf{60.21 \pm 22.68}$	**100**
	Greedy Closure	15.83 ± 8.52	$\mathbf{8.75 \pm 2.51}$	$\mathbf{1.25 \pm 0.55}$	1.85 ± 1.31	63.68 ± 24.41	**100**
	Greedy Min-Blockers	$\mathbf{15.83 \pm 8.51}$	9.10 ± 2.81	1.60 ± 0.94	3.00 ± 4.03	70.46 ± 34.66	**100**
	Random Slide	16.53 ± 8.81	9.04 ± 2.77	1.54 ± 0.85	2.81 ± 2.24	69.63 ± 26.45	**100**
	Random Push	82.49 ± 65.02	25.20 ± 14.73	7.66 ± 5.69	7.66 ± 5.69	109.58 ± 62.39	91.5

TABLE I: Simulation results. Values are means $\pm$ sample standard deviations across 20 scene-level means per category. Success aggregates 200 executions per method and category. Contact actions counts slides, or pushes for Random Push. Contact attempts counts their physical executions, including retries. T_p denotes planning time and T_e execution time. Bold marks the category-wise best values.

The slide is geometrically feasible when $K(i,d) \subseteq R$, giving the unit-cost rule $K(i,d) \to i$.

Exhaustive legal picks reach the least fixed point $\mathrm{cl}_P(R)$, independent of pick order. This follows because all preconditions are positive, so clearing further objects cannot disable an enabled rule. An optimal feasible plan can exhaust picks before each slide. Advancing a free pick preserves subsequent feasibility, and sliding an already pickable object can be replaced by its pick at lower cost.

For $S \subseteq V$, enable all pick rules and only slide-rule families targeting objects in S. Let $\Gamma(S)$ be their forward-chaining closure from $\varnothing$. A slide key satisfies $\Gamma(S) = V$, and

$$\mathrm{OPT}_S = \min_{S \subseteq V} \{|S| : \Gamma(S) = V\}. \tag{6}$$

A legal plan induces a key through its slid objects. Conversely, a successful derivation gives an executable sequence. A key can be verified in $O(L_{\mathcal{H}})$ time and space, where $L_{\mathcal{H}}$ is total rule incidence [17], [18].

Minimum-key selection is NP-complete in the abstract rule class, even with pick-rule bodies of size at most two. Membership follows from linear-time verification. Threshold-one/two Target Set Selection [22] reduces to free singleton/pair rules and unconditional paid rules for selected seeds, preserving the optimum. This result characterizes the abstract dependencies, without asserting that every such rule system has a cuboid realization.

D. Exact optimization and execution

For each object, let $\mathcal{A}_i$ contain its pick and slide rules, with preconditions $\mathrm{pre}(a) = B(g)$ or $K(i,d)$ and costs $c(a) = 0$ or 1. Rules dominated in cost and preconditions are removed. Binary variables x_{ia} select one rule per object, while continuous ranks u_i impose precedence:

$$\begin{aligned} \min \quad & \sum_i \sum_{a \in \mathcal{A}_i} c(a) x_{ia} \\ \text{s.t.} \quad & \sum_{a \in \mathcal{A}_i} x_{ia} = 1 && \forall i, \\ & u_j + 1 \le u_i + n(1 - x_{ia}) && \forall j \in \mathrm{pre}(a), \\ & x_{ia} \in \{0,1\}, \quad 0 \le u_i \le n-1. \end{aligned} \tag{7}$$

Selected rules require every blocker to precede its target, excluding cycles. Conversely, any legal sequence admits feasible ranks. A topological ordering therefore recovers an exact minimum-slide plan whenever the MILP is feasible.

The implementation solves the pruned MILP with HiGHS [23]. During execution, observations update object identities and poses, and the selected grasp is transformed to the current pose. Slides use top contact along the planned local direction. Reobservation determines whether further sliding is required before picking.

III. Experimental Evaluation

A. Benchmarks and evaluation protocol

We evaluate 40 heterogeneous scenes, equally divided into axis-aligned and non-axis-aligned configurations. The scenes vary object counts, dimensions, positions, contacts, and in-plane orientations. Each scene is executed 10 times per method, giving 2,000 Isaac Sim runs. Random Slide is additionally evaluated in 40,000 planner-only trials. Simulation uses a Franka Panda in Isaac Sim 5.0 at 120 Hz with rigid-body contact dynamics and continuous collision detection.

The proposed method minimizes planned slides. *Greedy Closure* exhausts picks and selects the feasible slide maximizing removals after subsequent pick closure. *Greedy Min-Blockers* minimizes the precomputed slide blocker count. *Random Slide* exhausts picks, then randomly selects a feasible object and slide direction. *Random Push* instead pushes a randomly selected approachable face at a deadlock, then observes and replans with the changed geometry.

We report total actions, planned slides, physical contact attempts including retries, execution time, and completion. These separate contact selection from its physical execution effort. Planning time T_p includes preprocessing and selection or optimization for the four offline methods, and cumulative replanning for Random Push.

B. Simulation results and scalability

Table I shows that the proposed method achieves the lowest or tied-lowest planned slide and total-action counts. Greedy Closure uses 0.10 additional slides in axis-aligned scenes and matches the proposed method in non-axis-aligned scenes.

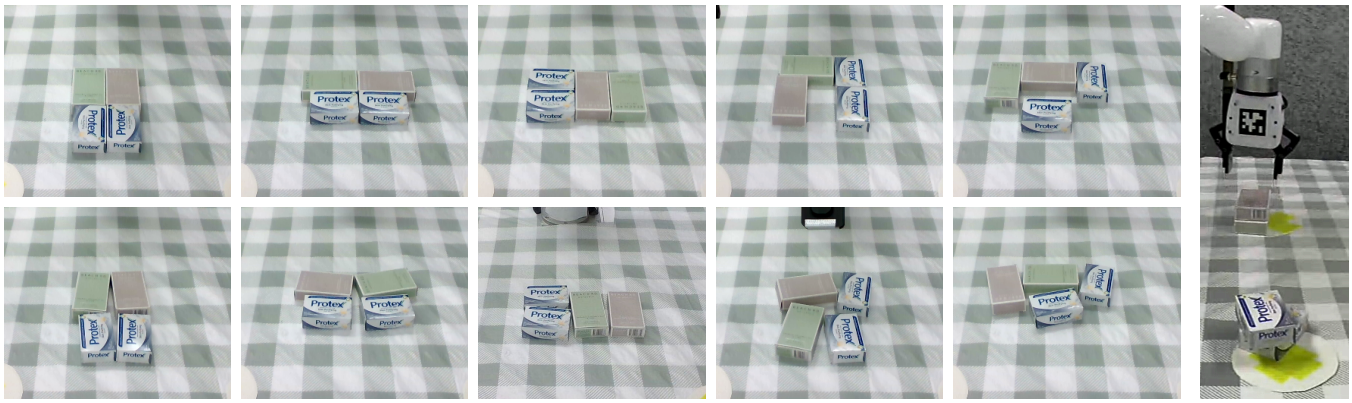


Fig. 2: Real-robot evaluation on five paired scenes. Axis-aligned layouts are shown in the top row and their non-axis-aligned counterparts in the bottom row. The task is to sort the two object classes to the designated destinations at right.

Category	T_p (ms)	Total actions	Slide actions	Slide attempts	T_e (s)	Success (%)
Axis-aligned	699.2 ± 32.6	5.20 ± 1.10	1.20 ± 1.10	1.80 ± 2.05	365.25 ± 115.45	100
Non-axis-aligned	727.0 ± 23.2	5.60 ± 0.55	1.60 ± 0.55	1.60 ± 0.55	348.31 ± 29.75	100

TABLE II: Real-robot results: mean $\pm$ standard deviation across five runs per category. T_p is the complete high-level planning phase. T_e includes observations, action execution, and the remaining robot workflow, excluding T_p. Slide attempts include retries.

Greedy Min-Blockers performs worse, as immediate blocker count does not capture subsequent pick opportunities. All four slide-based methods complete every run, allowing comparison of contact efficiency and execution effort.

The proposed method has the lowest reported mean contact-attempt counts and execution times in both categories. The 2.35 and 1.70 attempts exceed the corresponding 1.55 and 1.25 planned slides, reflecting reattempts after insufficient displacement. Random Push requires more actions and time, with completion rates of 65.5% and 91.5%. Failures occur when objects leave the robot's reach or new deadlocks lead to further pushes and timeout. These results emphasize the importance of selecting contact actions according to the grasp clearance they enable.

A separate scalability study uses 20 instances per category for each $n \in \{5, 10, 15, 20, 30, 40, 50\}$. All instances through $n = 40$ are solved to proven optimality. At $n = 50$, all 20 non-axis-aligned and 18 of 20 axis-aligned instances finish within 60 s. Median planning times increase from 9.01 ms to 3.88 s for axis-aligned scenes, and from 7.17 ms to 1.13 s for non-axis-aligned scenes, between $n = 5$ and $n = 50$.

C. Real-robot evaluation

The physical system uses a 7-DoF UFACTORY xArm7, an Intel RealSense D435i RGB-D camera, and an NVIDIA RTX 5090 workstation. SAM2 with seed-point initialization provides masks [24], and FoundationPose estimates 6D poses using its refiner and score networks [25]. Identity tracking maintains object associations across observations. Each scene contains two Beach boxes ($110 \times 65 \times 25$ mm) and two Protex boxes ($90 \times 60 \times 28$ mm), sorted to class-specific destinations.

Five scene geometries are evaluated in both categories, giving 10 physical trials (Fig. 2). Every non-axis-aligned scene requires at least one slide. Each run begins with perception and MILP planning, followed by the feedback-based execution described above.

All 10 runs complete successfully (Table II). Axis-aligned scenes average 1.20 planned slides and 1.80 attempts, including 0.60 retries. Non-axis-aligned scenes average 1.60 slides and 1.60 attempts. Mean execution times are 365.25 s and 348.31 s, respectively. Perception-guided reattempts accommodate insufficient displacement while retaining the planned sequence.

Physical planning times average 699.2 ms and 727.0 ms. These include conversion of perceived poses and meshes into planner geometry, whereas synthetic tests start from planner-ready geometry. Planning cost remains small relative to physical execution.

IV. Discussion and Conclusion

Optimality holds for the modeled geometric rules. Friction, compliance, approach feasibility, and controller behavior can require repeated contact attempts. Online perception assesses the outcome of sliding before grasping. The experiments support this integration on the tested scenes, while feedback ablations and systematic perturbation tests are needed to quantify robustness and recovery more broadly.

Monotonicity relies on irreversible clearance and fixed uncleared footprints. Stacking, neighbor displacement, or reintroduction of cleared objects requires additional state and may invalidate the dependencies. Local slide directions also limit feasibility. Richer preconditions, safety margins, and directional costs can extend the formulation under irreversible clearance. Applying the method to mobile manipulators or whole-body systems would additionally couple contact selection to reachability, balance, and force constraints.

The experiments demonstrate that exact geometric planning can guide contact selection within an integrated perception–planning–execution system. Incorporating the physical effort and uncertainty of these contact choices into the planning objective is an important direction for further work.

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